Review Questions for Lecture 13

1. Suppose you roll two dice, and let Z be the sum of the spots.

- What are the possible values of Z?
- What are the probabilities of each of the possible values?
- Make a probability histogram of Z.

2. Suppose you roll a die, and let X be the number of spots.
Let $Z = (X-3)^2$.

- What are the possible values of Z?
- What are the probabilities of each of the possible values?
- Make a probability histogram of Z.

3. Suppose you roll a die, and let X be the number of spots.
Suppose you independently flip a coin, and let $Y = 1$ if the coin lands heads, and $Y = 0$ if the coin lands tails. Let $Z = X + Y$.

- What are the possible values of Z?
- What are the probabilities of each of the possible values?
- Make a probability histogram of Z.

4. Suppose you roll a die, and let X be the number of spots.
Suppose you independently flip a coin, and let $Y = 1$ if the coin lands heads, and $Y = 0$ if the coin lands tails. Let $Z = XY$.

- What are the possible values of Z?
- What are the probabilities of each of the possible values?
- Make a probability histogram of Z.

5. Suppose you roll a die, and let X be the number of spots.
Suppose you independently flip a coin, and let $Y = 1$ if the coin lands heads, and $Y = 0$ if the coin lands tails. Let $Z = (X-3)^2(2Y-1)$.

- What are the possible values of Z?
- What are the probabilities of each of the possible values?
- Make a probability histogram of Z.

Questions 6-25 refer to the following scenario:
Suppose there are two boxes, A and B. Box A contains three tickets: 1,3,5. Box B has two tickets: 2,4. You draw one number from box A and one number from box B. Let X be the number from A, and Y be the number from B. Let

$$Z = XY, U = [X - E(X)][Y - E(Y)].$$

6. What are the possible values of X?

7. What are the possible values of $X + Y$?

8. What is $E(X)$?

9. What is $E(X + Y)$?

10. What is $SE(X)$?

11. What is $SE(X + Y)$?

12. What are the possible values of Y^2?

13. What is $P(Y^2 = 4)$?

14. What is $E(Y^2)$?

15. What is $SE(Y^2)$?

16. What is $E(Y)^2$?

17. What is $\sqrt{E(Y^2) - E(Y)^2}$.

18. What is $SE(Y)$?

19. What are the possible values of Z?

20. What is $P(Z = 12)$?

21. What is $E(Z)$?

22. What is $SE(Z)$?

23. What are the possible values of U?

24. What is $E(U)$?

25. What is $\mathrm{Cov}(X, Y)$?

26. Suppose that you randomly draw a number X from a box containing the numbers $-4,0,4$. What is $\mathrm{Cov}(X, X^2)$? Why is this example significant?

27. Suppose that $X_1, X_2, \ldots, X_n$ are independent, identically distributed random variables with expected value μ and $SE = \sigma$. Let

$$\bar{X} = (X_1 + X_2 + \cdots + X_n)/n$$

and define

$$V = \frac{(X_1 - \bar{X})^2 + (X_2 - \bar{X})^2 + \cdots + (X_n - \bar{X})^2}{n}.$$

Note that $E(X_1 + X_2 + \cdots + X_n) = E(X_1) + E(X_2) + \cdots + E(X_n)$.

a) Show that $E(V) = E(X_1^2) - E(\bar{X}^2)$.

b) Show that $E(X_i^2) = \sigma^2 + \mu^2$ for $i = 1,2, \dots, n$.

c) Show that $E(\bar{X}^2) = \mu^2 + \frac{\sigma^2}{n}$.

d) Show that $E(V) = \sigma^2 \left(\frac{n-1}{n}\right)$.

Note: This means that the variance of a list of numbers as we calculate it in class tends to underestimate the actual variance. The standard deviation that we calculate also tends to underestimate the actual standard deviation. This is why R represents the standard deviation by SD^+, dividing by $n - 1$ instead of n. This difference becomes less important when n is large.